COMP3670/6670
Introduction to Machine Learning
Week 8a: Dot Products, Orthonormal Bases & Orthogonal Projection

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Goals of this lecture

- Use dot products and cosine similarity to compare vectors.
- Recover coordinates directly from an orthonormal basis.
- Understand what orthogonal matrices preserve.
- Use Gram-Schmidt to construct orthonormal directions.
- Use orthogonal projection to find the closest point in a subspace.

- A basis gives coordinates for vectors in a subspace.
- If a datapoint lies in a k-dimensional subspace of $\mathbb{R}^d$, its coordinates need only k numbers, while k can be much less than d.
- A general basis requires solving a linear system to recover those coordinates.
- Orthonormal bases make coordinate recovery and projection simple.

Dot products measure alignment

Example 1. A query and two documents use term counts for the same three words:

$$\mathbf{q} = \begin{pmatrix} 4 \\ 1 \\ 3 \end{pmatrix}, \qquad \mathbf{d}_1 = \begin{pmatrix} 1 \\ 8 \\ 2 \end{pmatrix}, \qquad \mathbf{d}_2 = \begin{pmatrix} 6 \\ 2 \\ 5 \end{pmatrix}.$$

Which document is more aligned with the query?

Example 1 (cont'd). Multiply corresponding entries and add:

$$\begin{aligned} \langle \mathbf{q}, \mathbf{d}_1 \rangle &= 4 \times 1 + 1 \times 8 + 3 \times 2 = 18, \\ \langle \mathbf{q}, \mathbf{d}_2 \rangle &= 4 \times 6 + 1 \times 2 + 3 \times 5 = 41. \end{aligned}$$

Under raw dot-product scoring, document 2 ranks first.

It has larger counts for the words emphasized by the query.

Dot product. For $\mathbf{x}, \mathbf{y} \in \mathbb{R}^d$,

$$\langle \mathbf{x}, \mathbf{y} \rangle \;=\; \mathbf{x}^\mathsf{T}\mathbf{y} \;=\; \sum_{j=1}^{d} x_j y_j.$$

The dot product is symmetric and linear in each input.

In particular, $\langle \mathbf{x}, \mathbf{x} \rangle = \|\mathbf{x}\|^2$.

- Raw counts can favour a long document simply because it contains more words.
- For a query containing only “cat”, five mentions in 10,000 words beat two mentions in 100 words under a raw count score.
- Yet “cat” is more prominent in the shorter article.

  This motivates a *normalising* similarity score.

Cauchy-Schwarz inequality. For all $\mathbf{x}, \mathbf{y} \in \mathbb{R}^d$,

$$|\langle \mathbf{x}, \mathbf{y} \rangle| \leq \|\mathbf{x}\| \, \|\mathbf{y}\|.$$

Equality holds exactly when the vectors are linearly dependent.

This inequality implies a normalised dot product

$$\frac{\langle \mathbf{x}, \mathbf{y} \rangle}{\|\mathbf{x}\| \, \|\mathbf{y}\|}$$

is between -1 and 1 when $\mathbf{x}, \mathbf{y}$ are not zero vectors.

Proof idea. If $\mathbf{y} = \mathbf{0}$, the inequality is immediate. Otherwise set

$$a = \frac{\langle \mathbf{x}, \mathbf{y} \rangle}{\|\mathbf{y}\|^2}.$$

Since a squared norm is nonnegative,

$$0 \leq \|\mathbf{x} - a\mathbf{y}\|^2 = \|\mathbf{x}\|^2 - \frac{\langle \mathbf{x}, \mathbf{y} \rangle^2}{\|\mathbf{y}\|^2}.$$

Multiply by $\|\mathbf{y}\|^2$ and take square roots.

Angle and cosine similarity. For nonzero $\mathbf{x}, \mathbf{y}$,

$$\cos\theta = \frac{\langle \mathbf{x}, \mathbf{y} \rangle}{\|\mathbf{x}\| \, \|\mathbf{y}\|}, \qquad 0 \leq \theta \leq \pi.$$

The fraction is their cosine similarity.

For any vectors, $\mathbf{x} \perp \mathbf{y}$ means $\langle \mathbf{x}, \mathbf{y} \rangle = 0$.

The zero vector is orthogonal to every vector.



Example 2. Return to Example 1. The cosine similarities are

$$\frac{\langle \mathbf{q}, \mathbf{d}_1 \rangle}{\|\mathbf{q}\|\|\mathbf{d}_1\|} = \frac{18}{\sqrt{26}\sqrt{69}} \approx 0.425,$$

$$\frac{\langle \mathbf{q}, \mathbf{d}_2 \rangle}{\|\mathbf{q}\|\|\mathbf{d}_2\|} = \frac{41}{\sqrt{26}\sqrt{65}} \approx 0.997.$$

Document 2 also ranks first by cosine similarity: its vector points almost exactly in the query's direction.

Orthonormal bases make coordinates easy

Basis matrix. If $\mathcal{B} = (\mathbf{b}_1, \ldots, \mathbf{b}_k)$ is an ordered basis of $\mathcal{S} \subseteq \mathbb{R}^d$, put its vectors into columns:

$$\mathbf{B} = \begin{pmatrix} | & & | \\ \mathbf{b}_1 & \cdots & \mathbf{b}_k \\ | & & | \end{pmatrix} \in \mathbb{R}^{d \times k}.$$

For $\mathbf{x} \in \mathcal{S}$, we have $\mathbf{x} = \mathbf{B}[\mathbf{x}]_{\mathcal{B}}$.

The coordinate problem. To recover $[\mathbf{x}]_{\mathcal{B}}$ from $\mathbf{x}$, solve

$$\mathbf{B}\mathbf{y} = \mathbf{x}.$$

If $k = d$, the independent columns make $\mathbf{B}$ invertible, so

$$[\mathbf{x}]_{\mathcal{B}} = \mathbf{B}^{-1}\mathbf{x}.$$

An orthonormal basis makes this much easier: transpose replaces inverse.

Orthonormal vectors. The vectors $\mathbf{q}_1, \ldots, \mathbf{q}_k \in \mathbb{R}^d$ are orthonormal when

$$\|\mathbf{q}_i\| = 1 \quad \text{for every } i, \qquad \langle \mathbf{q}_i, \mathbf{q}_j \rangle = 0 \quad \text{when } i \neq j.$$

They have unit length and mutually perpendicular directions.

Proposition 1. Every orthonormal list is linearly independent.

Proof. Suppose

$$\alpha_1 \mathbf{q}_1 + \cdots + \alpha_k \mathbf{q}_k = \mathbf{0}.$$

Dot both sides with $\mathbf{q}_\ell$:

$$0 = \sum_{i=1}^{k} \alpha_i (\langle \mathbf{q}_\ell, \mathbf{q}_i \rangle) = \alpha_\ell.$$

This holds for every ℓ, so all coefficients are zero.

Orthonormal coordinates. Let $\mathbf{q}_1, \ldots, \mathbf{q}_k$ be an orthonormal basis of $\mathcal{S}$, and set

$$\mathbf{Q} = \begin{pmatrix} | & & | \\ \mathbf{q}_1 & \cdots & \mathbf{q}_k \\ | & & | \end{pmatrix} .$$

For every $\mathbf{x} \in \mathcal{S}$,

$$[\mathbf{x}]_{\mathcal{Q}} = \mathbf{Q}^{\mathsf{T}}\mathbf{x} = \begin{pmatrix} \langle \mathbf{q}_1, \mathbf{x} \rangle \\ \vdots \\ \langle \mathbf{q}_k, \mathbf{x} \rangle \end{pmatrix} .$$

Why the formula works. Write $\mathbf{x} = \sum_i \alpha_i \mathbf{q}_i$. For each j,

$$\langle \mathbf{q}_j, \mathbf{x} \rangle = \sum_i \alpha_i (\langle \mathbf{q}_j, \mathbf{q}_i \rangle) = \alpha_j.$$

Each coordinate is just one dot product.

No linear system or matrix inverse is needed.

Orthogonal matrix. A square matrix $\mathbf{Q} \in \mathbb{R}^{d \times d}$ is orthogonal if its columns are orthonormal. Equivalently,

$$\mathbf{Q}^\mathsf{T}\mathbf{Q} = \mathbf{I}_d, \qquad \mathbf{Q}^{-1} = \mathbf{Q}^\mathsf{T}.$$

For any $\mathbf{x}, \mathbf{y} \in \mathbb{R}^d$,

$$\langle \mathbf{Q}\mathbf{x}, \mathbf{Q}\mathbf{y} \rangle = \langle \mathbf{x}, \mathbf{y} \rangle, \qquad \|\mathbf{Q}\mathbf{x}\| = \|\mathbf{x}\|.$$

Thus orthogonal maps preserve lengths and angles.

Why dot products are preserved. Since $\mathbf{Q}^\mathsf{T}\mathbf{Q} = \mathbf{I}_d$,

$$\langle \mathbf{Q}\mathbf{x}, \mathbf{Q}\mathbf{y} \rangle = \mathbf{x}^\mathsf{T}\mathbf{Q}^\mathsf{T}\mathbf{Q}\mathbf{y} = \langle \mathbf{x}, \mathbf{y} \rangle .$$

Taking $\mathbf{y} = \mathbf{x}$ preserves squared lengths.

Cosine similarity therefore preserves angles between nonzero vectors.

Gram-Schmidt constructs orthonormal bases

Gram-Schmidt process. Start with linearly independent $\mathbf{v}_1, \ldots, \mathbf{v}_k$. Define

$$\mathbf{u}_1 = \mathbf{v}_1, \qquad \mathbf{q}_1 = \frac{\mathbf{u}_1}{\|\mathbf{u}_1\|},$$

and for $j = 2, \ldots, k$,

$$\mathbf{u}_j = \mathbf{v}_j - \sum_{i=1}^{j-1} (\langle \mathbf{q}_i, \mathbf{v}_j \rangle) \mathbf{q}_i, \qquad \mathbf{q}_j = \frac{\mathbf{u}_j}{\|\mathbf{u}_j\|}.$$

Then the $\mathbf{q}_i$ are orthonormal and span the same subspace as the $\mathbf{v}_i$.

Example 3. Start with three independent vectors in $\mathbb{R}^4$:

$$\mathbf{v}_1 = \begin{pmatrix} 2 \\ 4 \\ 4 \\ 8 \end{pmatrix}, \quad \mathbf{v}_2 = \begin{pmatrix} -10 \\ 10 \\ -20 \\ 20 \end{pmatrix}, \quad \mathbf{v}_3 = \begin{pmatrix} -28 \\ -16 \\ -11 \\ 58 \end{pmatrix}.$$

First $\|\mathbf{v}_1\| = 10$, so

$$\mathbf{q}_1 = \frac{\mathbf{v}_1}{10} = \frac{1}{5} \begin{pmatrix} 1 \\ 2 \\ 2 \\ 4 \end{pmatrix}.$$

Example 3 (cont'd). The component of $\mathbf{v}_2$ along $\mathbf{q}_1$ has coefficient

$$\langle \mathbf{q}_1, \mathbf{v}_2 \rangle = 10.$$

Subtract it to obtain the residual:

$$\mathbf{u}_2 = \mathbf{v}_2 - 10\mathbf{q}_1 = \begin{pmatrix} -12 \\ 6 \\ -24 \\ 12 \end{pmatrix}.$$

Example 3 (cont'd). The residual has squared norm

$$\|\mathbf{u}_2\|^2 = (-12)^2 + 6^2 + (-24)^2 + 12^2 = 900.$$

Thus $\|\mathbf{u}_2\| = 30$, and the second unit direction is

$$\mathbf{q}_2 = \frac{\mathbf{u}_2}{30} = \frac{1}{5}\begin{pmatrix} -2 \\ 1 \\ -4 \\ 2 \end{pmatrix}.$$

Example 3 (cont'd). The third vector has components along both directions:

$$\langle \mathbf{q}_1, \mathbf{v}_3 \rangle = 30, \qquad \langle \mathbf{q}_2, \mathbf{v}_3 \rangle = 40.$$

Remove both to obtain

$$\mathbf{u}_3 = \mathbf{v}_3 - 30\mathbf{q}_1 - 40\mathbf{q}_2 = \begin{pmatrix} -18 \\ -36 \\ 9 \\ 18 \end{pmatrix}.$$

Example 3 (cont'd). Since $\|\mathbf{u}_3\| = 45$,

$$\mathbf{q}_3 = \frac{\mathbf{u}_3}{45} = \frac{1}{5}\begin{pmatrix} -2 \\ -4 \\ 1 \\ 2 \end{pmatrix}.$$

The three $\mathbf{q}_i$ are orthonormal and span the same three-dimensional subspace as the $\mathbf{v}_i$.

Orthogonal projection extracts the explained component

- We have recovered coordinates for a vector already inside a subspace.
- A datapoint may lie outside the chosen low-dimensional subspace.
- Which point inside the subspace is closest to it?
- Projection keeps the explained component; the residual points perpendicular to every direction in the subspace.

Orthogonal projection. For a subspace $\mathcal{S} \subseteq \mathbb{R}^d$, a vector $\mathbf{p} \in \mathcal{S}$ is the orthogonal projection of $\mathbf{x}$ if

$$(\mathbf{x} - \mathbf{p}) \perp \mathbf{y} \qquad \text{for every } \mathbf{y} \in \mathcal{S}.$$

The vector $\mathbf{r} = \mathbf{x} - \mathbf{p}$ is the residual.

Projection theorem. Let $\mathbf{Q} \in \mathbb{R}^{d \times k}$ have orthonormal columns and $\mathcal{S} = \mathrm{Col}(\mathbf{Q})$. Then

$$\mathbf{p} \;=\; \mathbf{Q}\mathbf{Q}^{\mathsf{T}}\mathbf{x}$$

is the unique orthogonal projection of $\mathbf{x}$ onto $\mathcal{S}$.

It is also the unique closest point in $\mathcal{S}$ to $\mathbf{x}$.

Why the residual is orthogonal. Write

$$\mathbf{r} = \mathbf{x} - \mathbf{Q}\mathbf{Q}^{\mathsf{T}}\mathbf{x}.$$

Because $\mathbf{Q}^{\mathsf{T}}\mathbf{Q} = \mathbf{I}_k$,

$$\mathbf{Q}^{\mathsf{T}}\mathbf{r} = \mathbf{Q}^{\mathsf{T}}\mathbf{x} - \mathbf{Q}^{\mathsf{T}}\mathbf{Q}\mathbf{Q}^{\mathsf{T}}\mathbf{x} = \mathbf{0}.$$

So $\mathbf{r}$ is orthogonal to every column of $\mathbf{Q}$, hence every vector in $\mathcal{S}$.

Why projection is the nearest point. For any $\mathbf{y} \in \mathcal{S}$,

$$\mathbf{x} - \mathbf{y} = \mathbf{r} + (\mathbf{p} - \mathbf{y}),$$

where $\mathbf{r} \perp (\mathbf{p} - \mathbf{y})$. Therefore

$$\|\mathbf{x} - \mathbf{y}\|^2 = \|\mathbf{r}\|^2 + \|\mathbf{p} - \mathbf{y}\|^2 \geq \|\mathbf{r}\|^2.$$

Equality holds only when $\mathbf{y} = \mathbf{p}$.

Example 4. Let

$$\mathbf{Q} = \frac{1}{5}\begin{pmatrix} 4 & -2 & -2 \\ 2 & 4 & -1 \\ 2 & 1 & 4 \\ 1 & -2 & 2 \end{pmatrix}, \qquad \mathbf{x} = \begin{pmatrix} -8 \\ 9 \\ 3 \\ -4 \end{pmatrix}.$$

The three columns of $\mathbf{Q}$ are orthonormal. Find the closest point to $\mathbf{x}$ in their span.

Example 4 (cont'd). First compute the coordinates of the projection:

$$\mathbf{Q}^{\mathsf{T}}\mathbf{x} = \frac{1}{5}\begin{pmatrix} -12 \\ 63 \\ 11 \end{pmatrix}.$$

Then return to the original space:

$$\mathbf{p} = \mathbf{Q}\mathbf{Q}^{\mathsf{T}}\mathbf{x} = \frac{1}{25}\begin{pmatrix} -196 \\ 217 \\ 83 \\ -116 \end{pmatrix}.$$

Example 4 (cont'd). The residual is

$$\mathbf{r} = \mathbf{x} - \mathbf{p} = \frac{1}{25}\begin{pmatrix} -4 \\ 8 \\ -8 \\ 16 \end{pmatrix}.$$

Check that

$$\mathbf{Q}^{\mathsf{T}}\mathbf{r} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}.$$

Therefore $\mathbf{p}$ is the unique closest point in $\mathrm{Col}(\mathbf{Q})$.

Takeaways

- Dot products measure alignment; cosine similarity compares directions after normalisation.
- Coordinates in an orthonormal basis are dot products: $[\mathbf{x}]_{\mathcal{Q}} = \mathbf{Q}^{\mathsf{T}}\mathbf{x}$.
- Orthogonal matrices preserve dot products, lengths & angles.
- Gram-Schmidt turns linearly independent vectors into orthonormal directions.
- Projection $\mathbf{Q}\mathbf{Q}^{\mathsf{T}}\mathbf{x}$ is the nearest point in $\mathrm{Col}(\mathbf{Q})$.